

L10: Deffuant Model For Opinion or Consensus Formation

In some cases, it is a gross oversimplification to represent the state of each node as a binary variable (*active vs inactive*).

Instead, we need a scalar to represent the state of each node.

For example, our opinion about the risk of a potential COVID-19 infection may vary anywhere between the two extremes “*I am extremely worried*” to “*I do not care at all*”.

Typically, our opinion about such matters depends on the opinion of our social contacts.

The Deffuant model of opinion (*or consensus*) formation assumes that the state of a node v at time t is a scalar $s_v(t)$ that falls between 0 and 1.

A key parameter of the model is the “**tolerance threshold**” δ :

If the state of two connected nodes, say u and v , is greater than the threshold δ : $|s_v(t) - s_u(t)| \geq \delta$, the two neighbors “disagree” so strongly that they do not influence each other, and their state remains as is.

Otherwise, if $|s_v(t) - s_u(t)| < \delta$, then they influence each other, changing their state at the next time instant as follows:

$$s_v(t+1) = s_v(t) + \mu[s_u(t) - s_v(t)]$$

and

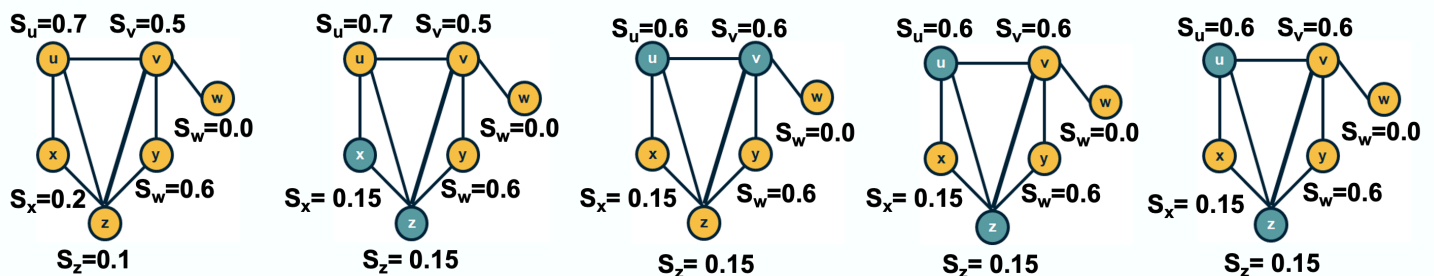
$$s_u(t+1) = s_u(t) - \mu[s_u(t) - s_v(t)]$$

where μ is the “**convergence**” parameter of the model.

If $\mu = 1/2$, then the state of the two nodes will become identical at time $t+1$. Typically μ is set to a value between 0 and $1/2$ to capture that the two neighbors may not reach complete agreement.

The model proceeds iteratively, by selecting a randomly chosen pair of neighbors in each iteration. If the network is weighted then the order at which pairs of nodes interact may depend on the weight of the corresponding edge.

Note that whether two neighbors influence each other or not may change over time.



For example, consider the network shown in the visualization ($\mu = 1/2, \delta = 0.45$). Suppose that we select pairs of neighboring nodes in the following order:

1. **x and z**: their opinion converges to $s_x = s_z = 0.15$
2. **u and v**: their opinion converges to $s_u = s_v = 0.6$
3. **u and z**: their opinion does not change
4. **v and y**: their opinion converges to $s_v = s_y = 0.6$

At that point, the opinion of every node has converged to a final value because any pair of adjacent nodes either have the same opinion, or their opinion differs by more than the threshold δ .

Food For Thought

The Deffuant model can result in interesting dynamics. Suppose that we have a social network in which almost all individuals are moderates (*say their initial opinion is close to 0.5*). There are also few extremist nodes though, some of them with an opinion that is close to 0 and others with an opinion that is close to 1.

Can you construct a scenario in which the entire network will become polarized, with every node being very close to one extreme or the other? What kind of initialization and network topology would make such an outcome more likely?